

COURSE 3 - DATA ENGINEERING 2026

C3-084A - PlantFabric Fabric Gate

Protected full assessment | fresh domain: manufacturing telemetry + production orders. Complete independently before opening its review.

Artifact	C3-084A
Status	2026 - LEARNER EDITION
Phase	C3-07 Cloud & Microsoft Fabric Data Engineering 15 lessons 50-70 focused hours
Prerequisite	C3-06 Gate PASS; C2B cloud foundations are recalled, while Fabric implementation is taught from first principles.
Current lane	Microsoft Fabric docs refreshed 2026-09-11 trial 60 days F4/F64 up to 1 TB OneLake
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House teaching rule: mental picture -> tiny example -> guided Fabric action -> expected result -> why it worked -> break/fix -> independent changed case -> validation -> security/production reasoning -> evidence. A portal screenshot is not mastery unless the underlying data, identity, lifecycle and recovery claim is independently verified.

Mission

PlantFabric needs a governed Fabric solution that moves batch data, handles incremental change, exposes a trusted analytical serving layer and operational event path, survives a controlled failure, and proves least-privilege access.

Required deliverables

- Architecture with storage/compute/connectivity/identity boundaries and justified Fabric item choices.
- Initial + incremental ingestion with stable keys/state, rerun convergence and reconciliation.
- Trusted Lakehouse contract plus relational serving query/view; event telemetry path with KQL-style operational query.
- Dev->Test change/promotion plan including dependency/reference remapping and post-deploy smoke check.
- Control-plane + data-plane security design with actual consumer-identity allowed/blocked test.
- Monitoring/recovery evidence plus capacity-aware operating note and handoff README.

Pass contract

PASS requires every required competency to PASS and zero unresolved critical failures. No aggregate score substitutes for a missing required competency.

Required competencies

- A - architecture/platform decisions
- B - movement/lakehouse/incremental correctness
- C - serving + event-query correctness
- D - lifecycle/change safety
- E - security/governance consumer proof
- F - observability/recovery
- G - capacity/operability
- H - reproducibility/handoff

Critical failures

- Invented Fabric runtime/security/monitoring evidence
- Credential/secret disclosure
- Wrong-grain or duplicate trusted state after incremental rerun
- Unsafe broad access used instead of required least privilege
- Security conclusion tested only as admin/privileged identity
- Claiming Git/deployment preserves or restores OneLake data
- Failure hidden/treated as success or destructive recovery
- Opening protected review before the attempt is saved

Runtime rule

A design-only submission can demonstrate architecture reasoning but cannot pass live-platform competencies. If Fabric access is unavailable, save the blocker honestly; do not fabricate completion.